

Relations

Definition (relation): A relation on a set A is a subset $R \subseteq A \times A$, abbreviated to $(x, y) \in R$ as $x R y$.

Example:

$$\text{Let } A = \{1, 2, 3, 4\}$$

$$\text{Then } R = \{(1, 1), (2, 1), (2, 2), (3, 3), (3, 2), (3, 1), (4, 4), (4, 3), (4, 2), (4, 1)\} \subseteq A \times A$$

so $1 R 1$ and $2 R 1$, but $1 \not R 2$ and $2 \not R 3$

Definition (relation properties):

- Relation R is reflexive if $x R x$ for every $x \in A$
- Relation R is symmetric if $x R y$ implies $y R x$ for all $x, y \in A$
- Relation R is transitive if whenever $x R y$ and $y R z$, then also $x R z$

Definition (equivalence relation): A relation R on a set A is an equivalence relation if it is reflexive, symmetric and transitive.

'=' is a good example of this:

- $\forall x, x = x$ (reflexive)
- $\forall x, y, x = y \Rightarrow y = x$ (symmetric)
- $\forall x, y, z, [x = y \wedge y = z] \Rightarrow x = z$ (transitive)

Definition (equivalence class): Suppose R is an equivalence relation on a set A . Given any element $a \in A$, the equivalence class containing a is the subset $\{x \in A: x R a\}$ of A consisting of all the elements of A that relate to a .

This is denoted as $[a]$. Thus the equivalence class containing a is the set $[a] = \{x \in A: x R a\}$

Consider the relation $R_1 = \{(-1, -1), (1, 1), (2, 2), (3, 3), (4, 4)\}$. The equivalence class 2 is the set $[2] = \{x \in A: x R_1 2\} = \{2, 2\}$.

For a different relation $R_2 = \{(-1, -1), (1, 1), (2, 2), (3, 3), (4, 4), (-1, 1), (1, -1), (-1, 3), (3, -1), (1, 3), (3, 1), (2, 4), (4, 2)\}$, the equivalence class containing 2 would be $[2] = \{x \in A: x R_2 2\} = \{2, 4\}$. Notice how $[4] = \{2, 4\}$, and so $[2] = [4]$.

Theorem: Suppose R is an equivalence relation on a set A . Suppose also that $a, b \in A$. Then $[a] = [b]$ if and only if $a R b$

Definition (partition): A partition of a set A is a set of non-empty subsets of A , such that the union of all the subsets equals A , and the intersection of any two different subsets is $\emptyset$.

Looking at the relation $\equiv (\text{mod } 3)$, we can see the equivalence classes:

$$[0] = \{3n\}$$

$$[1] = \{3n+1\}$$

$$[2] = \{3n+2\}$$

$$[3] = [0]$$

Thus forming 3 partitions of $\mathbb{Z}$: $[0]$, $[1]$, and $[2]$. We can call this $\mathbb{Z}_3 = \{[0], [1], [2]\}$

We can add and multiply the elements of $\mathbb{Z}_3$.

$$\begin{aligned} \text{Consider } [1] + [2] &\Rightarrow \{(3n+1) + (3m+2)\} \\ &= \{3(n+m)+3\} \\ &= \{3(n+m)\} \\ &= \{3r\} = [0] \end{aligned}$$

$$\text{Thus, } [1] + [2] = [0]$$

The same thing can be done with multiplication

Definition (integers modulo n): Let $n \in \mathbb{N}$. The equivalence classes of the equivalence relation $\equiv (\text{mod } n)$ are $[0], [1], [2], \dots, [n-1]$. The integers modulo n is the set $\mathbb{Z}_n = \{[0], [1], [2], \dots, [n-1]\}$. Elements of $\mathbb{Z}_n$ can be added by the rule $[a] + [b] = [a+b]$ and multiplied by the rule $[a] \cdot [b] = [ab]$.